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import cv2
import numpy as np
import matplotlib.pyplot as plt

# Load the image
# image = cv2.imread("image.jpg")

# if image is None:
#     print("Image not found!")
#     exit()

# print("Image loaded successfully!")
from google.colab import files
uploaded = files.upload()
image_name = next(iter(uploaded))
image = cv2.imread(image_name, cv2.IMREAD_GRAYSCALE)

if image is None:
    print("Error: Image could not be loaded.")
else:
    print("Image loaded successfully!")
    print("Image size:", image.shape)

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# Convert to grayscale
gray_image = image # The image was already loaded in grayscale

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# Contrast Stretching
min_pixel = np.min(gray_image)
max_pixel = np.max(gray_image)

contrast_stretched = ((gray_image - min_pixel) /
                      (max_pixel - min_pixel) * 255).astype(np.uint8)

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# Histogram Equalization
equalized_image = cv2.equalizeHist(gray_image)

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# CLAHE
clahe = cv2.createCLAHE(
    clipLimit=2.0,
    tileGridSize=(8, 8)
)

clahe_image = clahe.apply(gray_image)

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# Display Images
plt.figure(figsize=(5, 4))

plt.subplot(2, 2, 1)
plt.imshow(gray_image, cmap="gray")
plt.title("Original Grayscale")
plt.axis("off")

plt.subplot(2, 2, 2)
plt.imshow(contrast_stretched, cmap="gray")
plt.title("Contrast Stretched")
plt.axis("off")

plt.subplot(2, 2, 3)
plt.imshow(equalized_image, cmap="gray")
plt.title("Histogram Equalization")
plt.axis("off")

plt.subplot(2, 2, 4)
plt.imshow(clahe_image, cmap="gray")
plt.title("CLAHE")
plt.axis("off")

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plt.tight_layout()
plt.show()

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# Compare Histograms
plt.figure(figsize=(5, 4))

plt.subplot(2, 2, 1)
plt.hist(gray_image.ravel(), 256, [0, 256])
plt.title("Original Histogram")
plt.xlabel("Pixel Intensity")
plt.ylabel("Frequency")

plt.subplot(2, 2, 2)
plt.hist(contrast_stretched.ravel(), 256, [0, 256])
plt.title("Contrast Stretched Histogram")
plt.xlabel("Pixel Intensity")
plt.ylabel("Frequency")

plt.subplot(2, 2, 3)
plt.hist(equalized_image.ravel(), 256, [0, 256])
plt.title("Equalized Histogram")
plt.xlabel("Pixel Intensity")
plt.ylabel("Frequency")

plt.subplot(2, 2, 4)
plt.hist(clahe_image.ravel(), 256, [0, 256])
plt.title("CLAHE Histogram")
plt.xlabel("Pixel Intensity")
plt.ylabel("Frequency")

plt.tight_layout()

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# Save Enhanced Images
cv2.imwrite("contrast_stretched.jpg", contrast_stretched)
cv2.imwrite("histogram_equalized.jpg", equalized_image)
cv2.imwrite("clahe_image.jpg", clahe_image)

print("Enhanced images saved successfully!")

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plt.show()

#Observations

# | Technique | What to observe |
#| ----- | ----- |
#| Original | Baseline image with its original contrast |
#| Contrast Stretching | Wider intensity range and improved contrast |
#| Histogram Equalization | Redistribution of intensity values and enhanced overall contrast |
#| CLAHE | Local contrast enhancement, particularly useful when illumination varies |

# QUESTIONS AND ANSWERS:

# "" 1. What is image contrast, and why is it important in image processing?

# Answer:
# Image contrast is the difference in intensity between the dark and bright regions of an image. High-contrast images have clearly distinguishing features.
# Contrast is important because it improves the visibility of objects, edges, and details. Better contrast also helps computer vision systems perform better.

# 2. Explain the concept of an image histogram. What information does it provide?

# Answer:
# An image histogram is a graphical representation of the distribution of pixel intensity values in an image.
# For a grayscale image, intensity values normally range from 0 to 255, where 0 represents black and 255 represents white.

# A histogram provides information about:

# The distribution of brightness levels.
# Whether an image is dark or bright.
# The contrast of the image.
# The concentration of pixels within particular intensity ranges.

# It is useful for analyzing and improving image quality.

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3. Differentiate between Histogram Stretching and Histogram Equalization.

Answer:

Histogram Stretching Histogram Equalization

Expands the existing intensity range. Redistributes pixel intensities across the available range.

Improves contrast by mapping minimum and maximum values. Improves contrast using the cumulative distribution of pixel intensities.

Usually preserves the general shape of the histogram. Can significantly change the histogram distribution.

Simple and useful for low-contrast images. Useful when pixel intensities are concentrated in a limited range.

In this experiment, histogram stretching was implemented manually, while histogram equalization was performed using OpenCV.

4. What is Contrast Limited Adaptive Histogram Equalization (CLAHE)? How does it differ from standard Histogram Equalization?

Answer:

CLAHE stands for Contrast Limited Adaptive Histogram Equalization. It enhances contrast by dividing an image into small regions called tiles.

The main difference is:

Standard Histogram Equalization: Works globally on the entire image.

CLAHE: Works locally on different regions of the image.

CLAHE also limits excessive contrast enhancement using a clip limit, which helps prevent noise from being overly amplified.

Therefore, CLAHE is particularly useful for images having varying illumination or local contrast problems.

5. Why is histogram equalization commonly applied before feature extraction and image segmentation?

Answer:

Histogram equalization improves the distribution of pixel intensities and increases image contrast. This makes important features such as edges and textures more prominent.

As a result, feature extraction algorithms can identify useful features more effectively, while segmentation algorithms can better distinguish between different regions.

Therefore, histogram equalization can improve the quality of input data before subsequent computer vision operations.

6. Mention three real-world applications where histogram equalization is widely used.

Answer:

Three real-world applications are:

Medical imaging – improving the visibility of structures and details in medical images.

Satellite and remote-sensing images – enhancing features in satellite imagery.

Low-light photography – improving visibility and contrast in poorly illuminated photographs.

These applications are consistent with the types of real-world images suggested in the experiment.

7. What are the limitations of global histogram equalization?

Answer:

The main limitations of global histogram equalization are:

It considers the entire image rather than local regions.

It may over-enhance some areas.

It can amplify noise.

It may produce unnatural-looking images.

It may not perform well when different parts of an image have different illumination conditions.

CLAHE can help address some of these problems by performing enhancement locally and limiting excessive contrast enhancement.

8. How does contrast enhancement improve the performance of object detection and recognition systems?

Answer:

Contrast enhancement makes important image features such as edges, boundaries, textures, and objects more visible.

This can help object detection and recognition systems by:

Improving feature visibility.

Making object boundaries clearer.

Improving the distinction between objects and backgrounds.

Providing better input for feature extraction algorithms.

Therefore, contrast enhancement can make subsequent computer vision tasks more effective, especially when the original image has poor contrast.

9. Compare histogram-based enhancement techniques with brightness adjustment methods.

Answer:

Histogram-Based Enhancement Brightness Adjustment

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# Changes the distribution of pixel intensities. Mainly shifts pixel intensities toward brighter or darker values.
# Can improve image contrast. Primarily changes overall brightness.
# Includes histogram stretching, histogram equalization, and CLAHE. Usually involves adding or subtracting an intensity value.
# Can reveal details that are difficult to see in low-contrast images. May not improve contrast significantly.
# More suitable when intensity distribution needs improvement. Suitable when the main problem is excessive darkness or brightness.

# Thus, histogram-based techniques generally provide more control over contrast than simple brightness adjustment.

# 10. Why is CLAHE preferred for medical imaging and low-light image enhancement?

# Answer:
# CLAHE is preferred because it performs local contrast enhancement instead of applying the same enhancement to the entire image.

# This is useful for medical and low-light images because different regions may have different illumination levels. CLAHE can enhance details in low-contrast areas without overexposing the rest of the image.

# It also uses a clip limit, which helps reduce excessive amplification of noise.

# Therefore, CLAHE is particularly useful when an image contains varying illumination and important local details need to be enhanced.
# """

```

image.jpg
image.jpg(image/jpeg) - 49418 bytes, last modified: 8/20/2026 - 100% done
 Saving image.jpg to image (3).jpg
 Image loaded successfully!
 Image size: (510, 340)

Original Grayscale



Contrast Stretched



Histogram Equalization



CLAHE

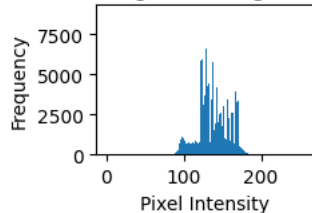


```

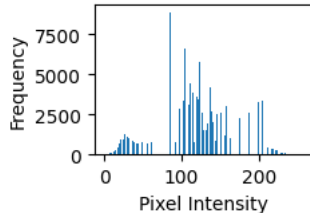
/tmp/ipykernel_2749/11686451.py:99: MatplotlibDeprecationWarning: Passing the range parameter of hist() positionally is deprecated since M
plt.hist(gray_image.ravel(), 256, [0, 256])
/tmp/ipykernel_2749/11686451.py:105: MatplotlibDeprecationWarning: Passing the range parameter of hist() positionally is deprecated since M
plt.hist(contrast_stretched.ravel(), 256, [0, 256])
/tmp/ipykernel_2749/11686451.py:111: MatplotlibDeprecationWarning: Passing the range parameter of hist() positionally is deprecated since M
plt.hist(equalized_image.ravel(), 256, [0, 256])
/tmp/ipykernel_2749/11686451.py:117: MatplotlibDeprecationWarning: Passing the range parameter of hist() positionally is deprecated since M
plt.hist(clahe_image.ravel(), 256, [0, 256])
Enhanced images saved successfully!

```

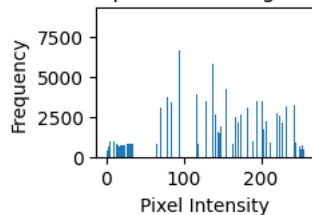
Original Histogram



Contrast Stretched Histogram



Equalized Histogram



CLAHE Histogram

